

Software-defined Radios (SDRs)

CUCUG
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May 21, 2026

What's a Radio?

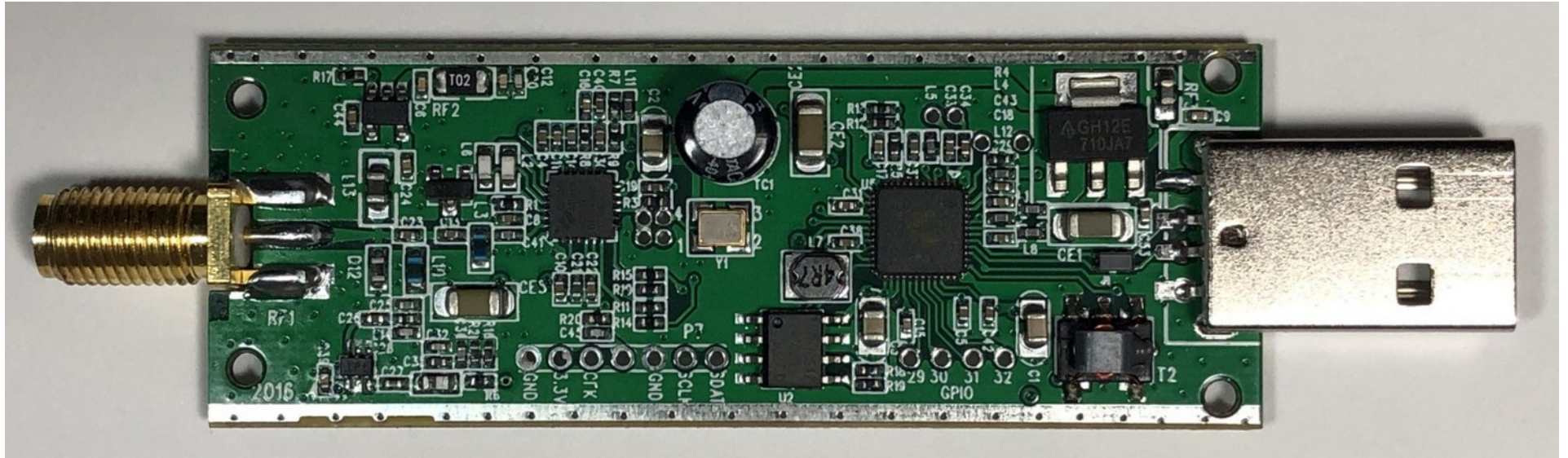
- Converts an EM spectrum signal into data or voice
- Antenna > Tuner > RF Amp > Demodulator > Audio amp > Output/speaker

What's a Software-defined Radio?



Well, almost. Might have a bit of filtering and amplifiers between stages.

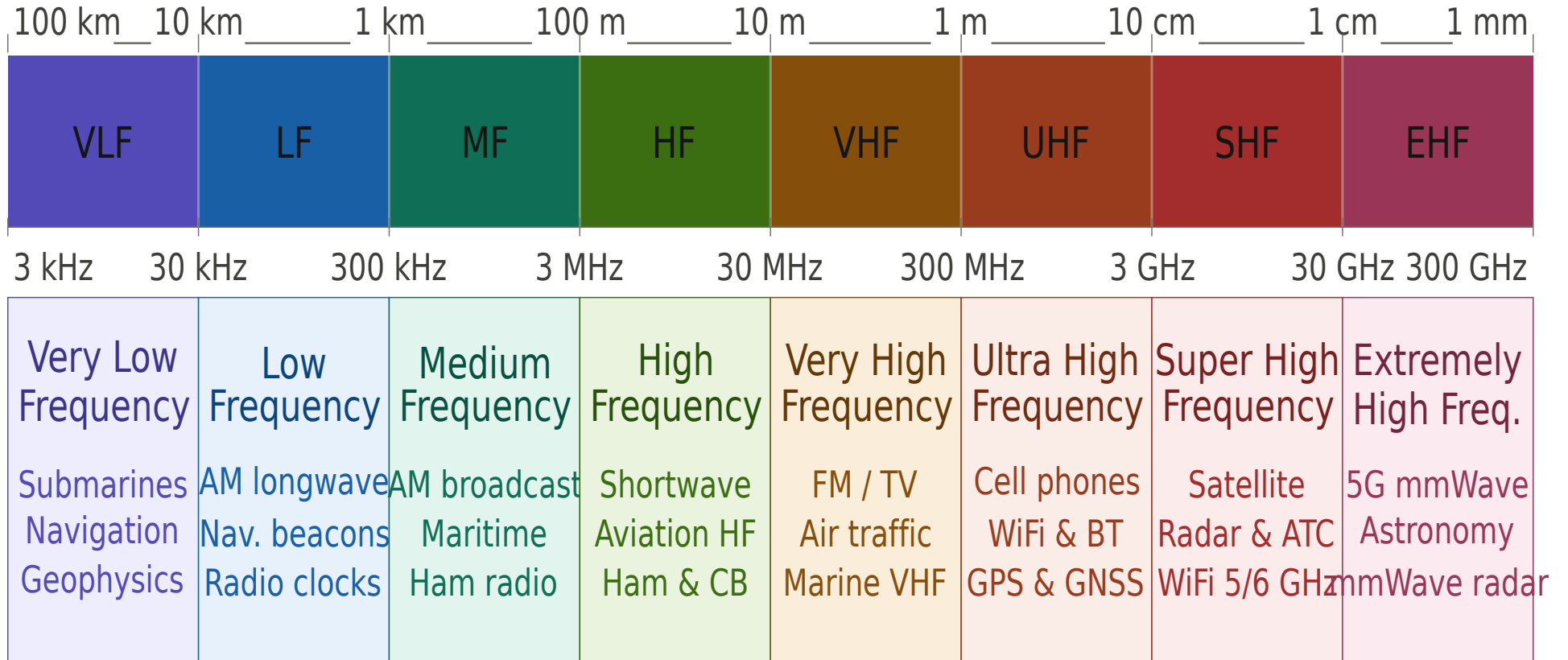
RTL-SDR v3 Guts



Units I Have Today

- RTL-SDR Blog v4
- Nooelec RTL-SDR v5
- Specs
 - 100 kHz to 1.75 GHz
 - ~2.4 MHz bandwidth
 - ADC: 8-bit
 - Interface: USB 2.0

Radio frequency spectrum



How Much?

- Mid-Tier (~\$100 - ~\$400)
 - [AirSpy](#)
 - [HackRF One](#)
 - [SDRplay](#)
- Budget (~\$35 - ~\$50)
 - [Nooelec NESDR SMArt v5](#)
 - [RTL-SDR Blog v3/v4/v4c/v4l](#)

But What Can I Do with It?

- Listen to just about any AM, FM, SW broadcast
- Capture [live pictures from satellites](#)
- Make a [trunked software scanner](#)
- Track airplanes overhead via their [ADS-B](#)
 - Log them on [FlightAware](#) or [FlightRadar24](#)
 - [Make Elon hate you!](#)
- Track [NWS weather balloons](#)
 - Log them on [Sondehub](#)

Thoughts to Keep in Mind

- Sensitivity vs. Selectivity
 - Block filters may be needed
- Bandwidth
 - SDR's "window" into the RF spectrum
- Analog-to-Digital Converter (ADC)
 - More bits, more signal resolution
- Antenna and placement is essential
 - Bigger is *not* always better!

Demo!